Status	Finished
Started	Tuesday, 27 October 2026, 4:51 PM
Completed	Tuesday, 27 October 2026, 4:56 PM
Duration	4 mins 57 secs
Grade	**13.00** out of 15.00 (**86.67**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

- a. Linear search ✓
- b. Binary search
- c. Hash search
- d. Both (1) & (3)

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

- a. It requires the list to be sorted ✓
- b. It is slower than linear search for small lists
- c. It can only be applied to large lists
- d. It does not work with negative numbers

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

- a. When the list is unsorted
- b. When the list is sorted ✓
- c. When the list contains duplicate elements
- d. When the list is very small

Question **4**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

- a. In the left sublist ✓
- b. At the beginning of the list
- c. In the right sublist
- d. In the entire list

Question **5**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

- a. Sequential search
- b. Binary search ✓
- c. Linear search
- d. Hash search

Question **6**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

- a. When the list is small and unsorted ✓
- b. When the list is large and unsorted
- c. When the list is large and sorted
- d. When the list is small and sorted

Question **7**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

- a. The search is considered unsuccessful ✓
- b. The last element is returned
- c. An error is raised
- d. The first element is returned

Question **8**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the list contains duplicate elements
b. When the search algorithm finishes
c. When the element is found in the list ✓
d. When the list is sorted

Question **9**

Incorrect

Mark 0.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. There must be a mechanism to delete and/or insert elements in the list
b. The list must be sorted
c. There should be direct access to the middle element in any sublist
d. Number values should only be present ✗

Number values should only be present

Question **10**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n/2
b. n ✓
c. n-1
d. log n

Question **11**

Incorrect

Mark 0.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 89 and 94
b. 90 and 99
c. 89 and 99
d. 90 and 94 ✗

Question **12**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

- a. O(log n) ✓
- b. O(n log n)
- c. O👎
- d. O(1)

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

- a. Quick search
- b. Merge search
- c. Linear search ✓
- d. Bubble search

Question **14**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

- a. O👎
- b. O(n log n)
- c. O(1) ✓
- d. O(log n)

Question **15**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

- a. Linear search requires the list to be sorted.
- b. Linear search divides the list into halves.
- c. Linear search is more efficient than binary search on large lists.
- d. Linear search can be applied to both sorted and unsorted lists. ✓